



Figure 1.6.5. A vortex panel on the surface of an arbitrary airfoil section.

The normal velocity at each control point, induced by all $n-1$ panels and the uniform flow, must be zero. This gives $n-1$ equations for the n unknown nodal vortex strengths.

For the remaining equation, we know that the flow must leave the airfoil from the trailing edge. This means that the velocity just above the trailing edge must equal the velocity just below the trailing edge. If the angle between the upper and lower surfaces at the trailing edge is nonzero, the trailing edge is a stagnation point. If the angle between the upper and lower surfaces at the trailing edge is zero, the velocity at the trailing edge can be finite but the velocity must be continuous across the trailing edge. Since there is discontinuity in tangential velocity across a vortex sheet, we know that at the trailing edge, the discontinuity across the upper surface is equal and opposite to the discontinuity across the lower surface. Thus, the net discontinuity across both surfaces will be zero. Because the discontinuity in tangential velocity across any vortex sheet is equal to the local strength of the sheet, at the trailing edge, the strength of the upper surface must be exactly equal and opposite to the strength of the lower surface. That is,

$$\gamma_1 + \gamma_n = 0$$

This is called the *Kutta condition* and it provides the remaining equation necessary to solve for the n unknown nodal strengths.

The 2×2 panel coefficient matrix in airfoil coordinates, $[\mathbf{P}]_{j(x,y)}$, for the velocity induced at the arbitrary point (x,y) by panel j , extending from the node at (x_j, y_j) to the node at (x_{j+1}, y_{j+1}) , is computed from the algorithm

$$l_j = \sqrt{(x_{j+1} - x_j)^2 + (y_{j+1} - y_j)^2} \quad (1.6.19)$$

$$\begin{Bmatrix} \xi \\ \eta \end{Bmatrix} = \frac{1}{l_j} \begin{bmatrix} (x_{j+1} - x_j) & (y_{j+1} - y_j) \\ -(y_{j+1} - y_j) & (x_{j+1} - x_j) \end{bmatrix} \begin{Bmatrix} (x - x_j) \\ (y - y_j) \end{Bmatrix} \quad (1.6.20)$$

$$\Phi = \text{atan2}(\eta l_j, \eta^2 + \xi^2 - \xi l_j) \quad (1.6.21)$$

$$\Psi = \frac{1}{2} \ln \left[\frac{\xi^2 + \eta^2}{(\xi - l_j)^2 + \eta^2} \right] \quad (1.6.22)$$